Homework Assignment Week 8

Objective:

- To be able to read data from a text file
- To be able to store the data in a list
- To be able to use the list methods for calculations
- To be able to handle the exceptions in the code

Instructions:

1. Name your python file as HW_W8_{yourFirstName}{LastNameInitial}.py
   For example: Michael K will name the python file of homework assignment Week 8 as **HW_W8_MichaelK.py**
2. The code should only use **Python commands that have been covered in the course up to this point.**
3. You will record a small video (3-5 min) to explain the code you wrote for the assignment.
4. Submit **both, .py file and the video**, in D2L Assignments -> Homework Assignment Week 8 Dropbox.

Grading Rubric:

- Detailed rubric for the assignment is given in the D2L Assignments -> Homework Assignment Week 8
- 15 points are assigned for the Python file submitted
- 10 points are assigned for the video presentation

Problem: Sales data analysis

Write a program that reads the sales data for a month from text file named `sales_data.txt` and stores it in a list. The text file is provided in D2L's Homework Assignment Week 8. Then it should calculate the following data: [1] Lowest sales in the month [2] Highest sales in the month [3] Total sales [4] Average sales. Finally, it should display the sales values read from the text file, lowest sales in the month, highest sales in the month, total sales of the month and average sales of the month.

Program should be written to handle following exceptions:

- It should handle any `IOError` exceptions that are raised when the file is opened and data is read from it.
- It should handle any `ValueError` exception that are raised when the item is read from the file and is converted to a number. This exception should print the error message when a particular sales value is not in correct format but continue to read other sales values from the text file.

Expected Output:

Test case 1: Output when no error

```
Minimum sales value= $950.0
Maximum sales value= $6321.0

Total sales= $92,281.00

Average sales value=$3,076.03
```

Test case 2: Handle exception when file cannot be opened

Steps for this test case: Delete the sales_data.txt from the folder which has HW Week 8 python file. Run the python file.

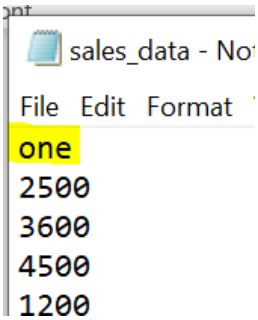
Expected output:

```
  File c:\users\og2360lu\aa normandale college\aa intro to
python\week8\assessment\homework\hw6_solution.py:26 in main
    raise IOError("An error occured while opening the file")

OSError: An error occured while opening the file
```

Test case 3: Handle exception when value cannot be converted to float

Steps for this test case: Put back the sales_data.txt in the folder which has HW Week 8 python file. Open the text file and add a text line as shown below and save the text file. Run the python file.



Expected output:

```
Warning: Could not convert 'one' to a number. Skipped.

Minimum sales value= $950.0
Maximum sales value= $6321.0

Total sales= $92,281.00

Average sales value=$3,076.03
```